[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

DITCHING MACHINES, BOOM TYPE; CRAWLER-MOUNTED, DED

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers commercial model diesel-engine-driven (DED), self-propelled, boom type, crawler mounted ditching machine. The ditching machines are intended for use such as ditching, excavating for building foundations, drainage, and pipeline ditches at federal facilities. This CID covers only one classification of ditching machines.

2. SALIENT CHARACTERISTICS.

2.1 Description. The ditching machine consist of a power plant, power train, digging boom and chain, buckets or cutters or both, spoil conveyor, framework, and crawlers.

2.2 Standard commercial product. The ditching machines shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this commercial item description but which are a part of the manufacturer's standard commercial product, shall be included in the ditching machines being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301.

AMSC N/A FSC 3805

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2.3 Safety. Rotating or reciprocating parts and parts subject to high operational temperatures, that are of a nature or are located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or guarded. The ditching machine shall have Rollover Protection Structure (ROPS) to conform with the Society of Automotive Engineers SAE J 1040 and 29 CFR 1926.1000 and comply with applicable Occupational Safety and Health Administration (OSHA) regulations in effect at time of manufacture.

2.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

2.5 Operational requirements. The ditching machine shall be capable of operating satisfactorily in ambient temperatures from 0 degrees Fahrenheit (°F) up to 125 °F (- 18 degrees Celsius (°C)) up to 52 °C).

2.6 Performance. The ditching machine shall be capable of digging into various earth formations a ditch up to 24 inches (609.6 millimetre (mm)) in width, and:

a. Not less than 6 feet (1.8 metre (m)) deep in firm clay soil at a digging rate of not less than 5 feet per minute (ft/min) (25.4 m per second (m/s));

b. Not less than 8 feet (2.4 m) deep in clay-loam soil at a digging rate of not less than 3 ft/min (15.2 m/s) or;

c. Not less than 6 feet (1.8 m) deep in clay-gravel soil at a digging rate of not less than 4 ft/min (20.3 m/s).

In addition to the above, the digging machine, when equipped with special digging teeth (see 2.17) shall also be capable of digging coral, frozen, or hard ground at a slower rate of speed. The crowding speed shall be individually selected by the operator. The discharge conveyor shall be capable of depositing spoil at a distance sufficient to prevent reentry of spoil into the trench. The ditching machine shall have a transporting height of not more than 146 inches (3 708 mm) with boom not less than 6 inches (152 mm) above ground.

2.6.1 Travel speeds. The ditching machine shall have hydrostatic track drive with infinitely variable forward and reverse speeds or three or more travel speeds forward and one reverse.

2.6.2. Digging speeds. The ditching machine shall have a continuously variable digging (crowding) speeds from 0 to not less than 6 ft/min (30.5 m/s) at engine governed speed.

2.7 Digging boom. The boom shall be provided with a safe, accessible and adequate method of adjusting the digging chain. Hydraulic cylinders for lifting the boom must provide enough down pressure to support rear of unit. A hydraulic hoist, operating independently of the other drives, shall be provided to enable the operator to accurately regulate the digging depth and to raise the boom into travel position.

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2.8 Digging chain. The digging chain shall be constructed so each bucket or cutter may be removed individually. An overload safety release between the driving mechanism and the digging chain shall be provided. The digging chain shall have means for reversing the direction of travel for cleaning purposes. Means for adjusting the digging chain shall be provided.

2.9 Buckets. When digging buckets are provided, they shall be of the self-cleaning design which will remove the spoil from the bucket either by centrifugal force, snap action, or positive action of a wiper plate. The ditching machine shall be equipped with buckets and teeth to cut trenches, one at a width of not less than 18 inches (457.2 mm), and one at a width of not less than 24 inches (609.6 mm). The 24-inch (609.6 mm) cutting width requirement may be met with bolted-on side cutters. The teeth shall be of the quick interchangeable insert type. Replacement of teeth shall be accomplished without the use of special tools.

2.10 Cutters. When digging cutters are provided, they shall cut trenches, one at a width of not less than 18 inches (457.2 mm), and one at a width of not less than 24 inches (609.6 mm). The cutters shall be of the interchangeable type.

2.11 Cleanup scraper. Vertical boom type ditching machine, when furnished, shall be equipped with a cleanup scraper (crumber) which will effectively clean loose spoil spillage from the bottom of the trench. The scraper shall have an automatic trip or hydraulic fold-up option from the operator's location to release obstacles.

2.12 Spoil conveyor. The spoil conveyor shall be individually controlled, dischargeable to either side, from the operator's seat, and belt speed so controlled as to provide the correct operational speed to give maximum or minimum discharge. Means shall be provided to align conveyor and pulleys at right angles with center line of belt, when end pulleys are used for tension adjustment.

2.13 Spoil conveyor belt. The spoil conveyor belt shall be adequate width to handle the spoil while digging at maximum depth and maximum crowding speed. The carrying side of the belt shall be either of the cleat or trough type surface.

2.14 Crawlers track. The crawlers track shall have a smooth faced, forged, or rolled steel semigrouser, or grouser type track shoe. Ground bearing area of the crawler tread shall be adequate to ensure a bearing pressure of not more than 10 pounds per square inch gage (psig) (68.9 kilopascal (kPa) (gage)). Provisions shall be made to adjust for slack due to wear.

2.15 Engine. The ditching machine shall be powered by the manufacturer's standard commercial engine normally furnished to the commercial market for this size ditching machine. The engine shall be furnished with an electrical cranking system. The engine shall be furnished with accessories normally equipped on the manufacturer's commercial standard ditching machine.

2.16 Controls. All controls shall be located within easy reach of the operator and shall be clearly marked as to use and function.

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2.17 Special cutters. When specified (see 6.2), the ditching machine shall be supplied with one set of special cutters for each digging width (see 2.9 and 2.10). The special cutters, which shall facilitate digging in coral, hardpan, or frozen ground, shall be in addition to the buckets or cutters normally supplied.

2.18 Lifting and tiedown attachments. The ditching machines shall be equipped with lifting and tiedown attachments. A nonferrous transportation plate shall be provided and mechanically attached to the ditching machines. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the ditching machines on the carrier when shipped.

2.19 Toolbox. When specified (see 6.2), the ditching machine shall be provided with a lockable metallic toolbox securely fastened to the ditching machine in a protected and accessible location.

2.20 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J 534. Where use of high pressure lubricating equipment, 1,000 psig (6 894 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with the type of lubricant specified in the owner's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

2.21 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.0635 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

2.22 Electromagnetic radiation. Electromagnetic radiation from the ditching machine shall be not greater than the limits recommended in SAE J 551.

2.23 Identification marking. Identification shall be permanently and legibly marked directly on the ditching machines or on a corrosion-resisting metal plate securely attached to the ditching machines at the source of manufacture. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

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2.24 Instruction plate. The ditching machines shall be equipped with instruction plates or decals, suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, rivets, or bolts of not less than 0.125-inch (3.175 mm) diameter.

2.25 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10, and all other requirements of this commercial item description including form, fit and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS.

3.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 The Federal Acquisition Regulation (FAR), and OSHA standards may be obtained from the Superintendent of Documents, U.S. Printing Office, Washington, DC 20402.

6.1.2 ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.1.3 SAE Standards are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. When special cutters are required (see 2.17).
c. When a toolbox is required (see 2.19).
d. Color of finish coat (see 2.21).

6.3 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	NOMENCLATURE
3805-00-089-2192	Ditching Machine

6.4 Supersession data. This commercial item description replaces Federal Specification OO-D-2747, dated March 23, 1991.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

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